# Chapter 13

## System Operation and Maintenance

# System Operation

- Meaning
  - System in operation until it cannot provide any useful functions to the users anymore.
- Purpose
  - Support Users in using the system to help them perform their works.
- Sources of support
  - In-house and Vendors
- Forms of support
  - User training, Help desks, Online and Automated support

# <u>Forms</u> of Operation Supports

# 1. User Training

- Teach users how to use the system
- Support users when they face problems using the system.
- System undergone changes or enhancement
  - If changes are major – use formal training
  - If changes are minor – use e-mail support, training manual supplements, user guides, online help facilities and special training Web site.

# 2. Help Desks

- Help the users in various ways :
  - Answer operational or technical questions.
  - Teach users how to use the system to meet their information needs
  - Teach users how to use system resources in more effective manner
- Differentiated from Maintenance
  - Help desk - provide assistance to the users to be more effective in performance of their jobs and lead the users solving problems for themselves
  - Maintenance - aims to provide continuous and uninterrupted supply of computer resources to the users.
- Help desks personnel. Strong technical and interpersonal skills, solid understanding of the business, well trained for the new system

# 3. Automated Supports

- **Purpose** : more supports moved online and automated to cut down manual cost of providing support.
- **Examples** : e-mail responses, FAQs, bulletin boards, automated voice mail, and online help facilities.
- **On-line support forums**. Allow users access to information on bugs, their new product releases, and tips for more effective usage of system. Online services over company intranets.

# 3. Automated Supports

- Online chat rooms. Eg. Blackboard, popularly used in universities and colleges to help deliver online courses and to allow students and lecturers to meet online.
- Online knowledge. Technical and support information about vendor products and information for on-site personnel to use in solving problems. Some vendors allow users to access some of these information.
- Voice-response systems. Allow users to search for some pre-recorded messages on various issues, such as, usage, problems and solutions. These messages are also called podcasts.

# Examples of Operation Support

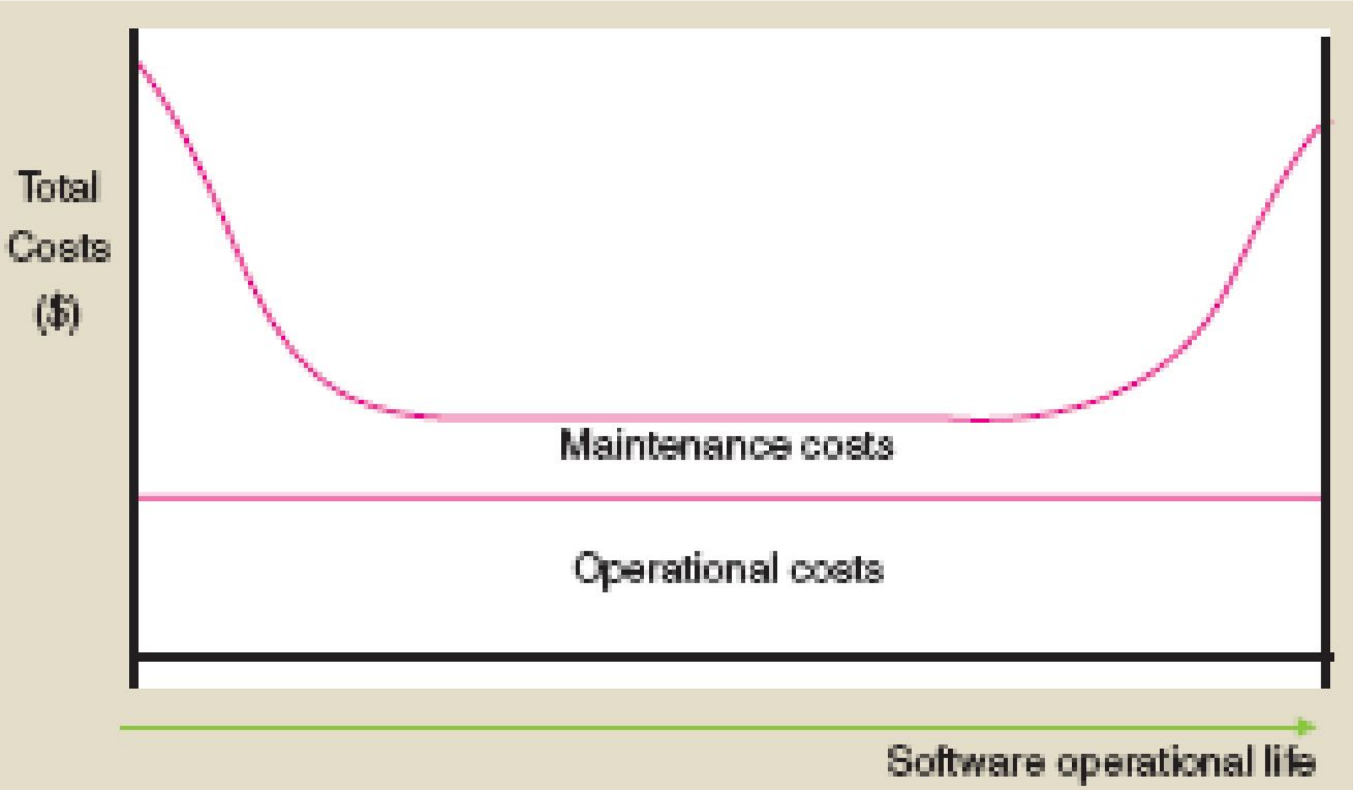
1. Assist users in writing simple programs in end-user computing
2. Install new hardware or software for the users
3. Assist users in evaluating a particular software
4. Help users to submit system proposal
5. Teach users (new and old) how to use the system
6. Help users to solve problems encountered while using the system
7. Assist users in extract data from server databases onto their personal computers
8. Set up new user accounts for a particular system

# System Maintenance

# What is Maintenance ?

- Operational life
  - Begins : System is implemented
  - Ends : System no longer useful eg. not economical, replaced etc
- Operation Vs Maintenance
  - System maintenance □ IT Personnel
  - System operation □ End-users
- Purposes : to ensure that the system will continue to be functioning at an acceptable level to the users by improving or modifying the system.

# Costs of Maintenance




- **Diagram shows :**
  - **Operational cost – quite constant**
  - **Maintenance cost – varies in the shape of ‘bath-tub’ curve**

# 3 Types / Purposes of Maintenance

# 1. Perfective (Users Requests)

- **Meaning :**
  - To cope with new requirements due to users' requests (eg. new markets, new information required, new technologies)
  - The requests by users may include to improve software functionality, usability and performance.
- **Examples**
  - Improving performance eg. faster response to customer service as requested by users.
  - Improving functionality eg to add new features or reports as requested by users.
  - Improving usability eg improving layouts of some reports in an inventory system as requested by users.
  - Improving usability eg improving the interface (to make

# 2. Corrective (Correction of Errors)

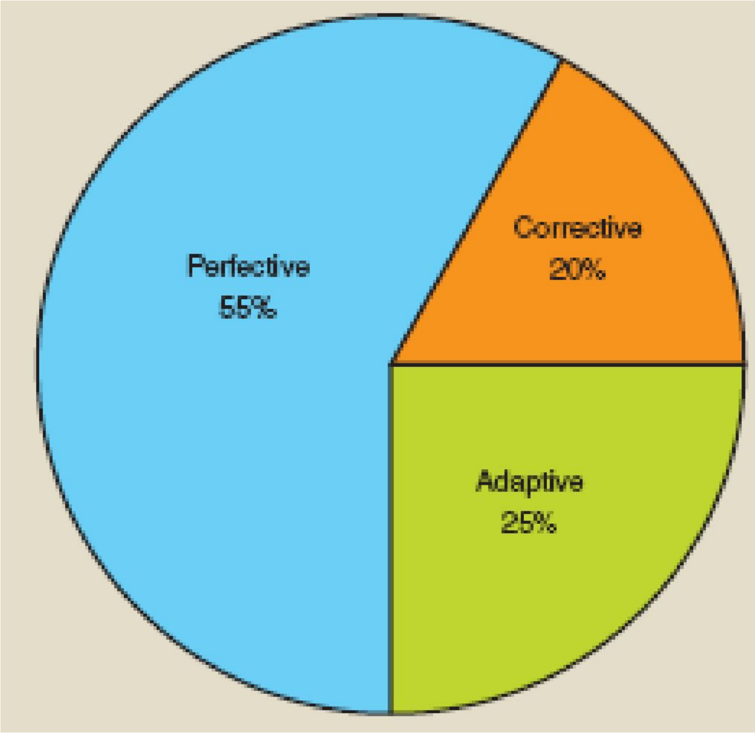
- **Meaning**
  - To fix bugs or faults (in programs) which are unforeseen and arise during operation.
  - To keep the programs working correctly.
- **Examples**
  - Diagnose and fix logic errors or program code.
  - Update software drivers
  - Restore proper configuration settings

# 3. Adaptive (Changes in Environment)

- **Meaning**
  - To **adapt** the system due to changing environment :
    - **<u>business</u> environment** (eg. new regulatory requirements)
    - **<u>computer</u> environment** (eg. hardware or software changes such as a change of OS)
- **Examples**
  - Changes to payroll system due to new employment *laws.*
  - Add *new online capability* due to business environment changes
  - Create *new reports* due to business environment changes
  - Add *new data entry field* to input screen due to business environment changes
  - New systems will be added eg. OS, hardware, DBMS

# Distribution Costs Of Maintenance

- **Maintenance costs is significantly very high up to up 70% of the total life cycle costs.**
- **Percentage distribution of maintenance**



# Reducing Maintenance Costs

- Thorough Analysis of Requirements. If systems can be correctly and fully specified and agreed there will be little need for maintenance.
- Predictive Design. Any likely changes in requirements could be built into the software at design stage thus eliminating or reducing the need for maintenance.
- Decompositon of Program (modularity). This can help maintenance. It is easier to make changes to small programs than larger ones.

# Reducing Maintenance Costs

- Thorough Program Testing. This would reduce the incidence of errors in the programs during operation.
- Complete Documentation. Up-to-date documentation would permit changes to be made easily and efficiently.
- Usage of Methodology and Productive Tools. Efficient methodologies, such as prototyping and structured approach and use of CASE tools can help to carry out maintenance.

# Exercise for Students

- **Identify an appropriate type of maintenance which each of the following can reduce :**
  - Conduct thorough program testing
  - Carry out predictive design taking into accounting potential features required.
  - Keep complete and up-to-date documentation for the system.
  - Design the system to be more modular.
  - Conduct thorough analysis of requirements.
  - Use efficient methodology and productive tools.

# Suggested Answers

- Conduct thorough program testing. (perfective)
- Carry out predictive design taking into accounting potential features required. (perfective)
- Keep complete and up-to-date documentation for the system. (all types)
- Design the system to be more modular. (all types)
- Conduct thorough analysis of requirements. (perfective)
- Use efficient methodology and productive tools. (all types)

# Software Maintenance Contract

- Meaning
  - Parties : Between the customer and the software supplier
  - Fees : Customer pays an annual fee (usually some percentage of the original software cost).
  - Time : Specific timeframe and renewable.
- Content or Services
  - Information. Extra information (eg. Factsheets, newsletter) about using the package
  - Help. Helpline facility allows users to resolve problems
  - Updates. Fixing software faults emerged.
  - Upgrades. New version at discounted price.
  - Legal conditions eg. Duration, customers’

# User Groups

- A user group is a forum for users of particular hardware or, more usually, software, so that they can share ideas and experience and, on occasions, acting as an arbiter (intermediary) in disputes with supplier.

- Ways to Set up
  - By Software Manufacturers. Software manufacturer use them to maintain contact with customers and as a source of new product ideas.
  - By Groups of Users. Users who were not satisfied with the level of support they were getting from suppliers of proprietary software.

# User Groups

- Benefits to Users
  - Share ideas. Users of a particular package can meet, or perhaps exchange views over the Internet to discuss solutions, ideas or ‘short cuts’ to improve productivity.
  - Feedbacks. Ideas from members and IT specialists can be incorporated into future system development.
  - Arbiter. Acts as an arbiter in any disputes between users and suppliers eg functionality disputes.

# Examples of User Groups

- Hardware platforms
  - 3Com Users Group (3CUG)
  - IBM mainframe - SHARE
  - Boston Mac Users Group
- Software platforms
  - Adobe User Group - Philippines
  - International Informix Users Group (IIUG)
  - UK Oracle User Group
  - ASUG (Americas’ SAP Users’ Group)

# Example


Oracle User Group Community
A worldwide community of independent user groups.
Learn More or Join Today